

ARTIFICIAL INTELLIGENCE

Lecture Set 10

**Uncertainty, Conditional Probability, Bayes Rule, Naïve Bayes Classifier,
Bayesian Networks**

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Uncertainty

- Uncertainty refers to a lack of certainty or knowledge about future events or outcomes.
- It arises when information is incomplete, imprecise, or ambiguous, making it difficult to predict with absolute certainty what will happen.
- Probability theory provides a framework for quantifying and managing uncertainty.
- By assigning probabilities to different possible outcomes, we can express our degree of belief or confidence in those outcomes.
- Probability allows us to make informed decisions and predictions even in the presence of uncertainty, helping us assess risks, allocate resources, and make optimal choices based on available information.

Probability

1. **Probability:** Probability is a **measure of the likelihood** of an event occurring. It is a numerical value between 0 and 1, where 0 indicates impossibility and 1 indicates certainty.
2. **Prior Probability:** Prior probability represents your **initial belief** or understanding about the likelihood of an event occurring **before any evidence** or observation is taken into account.
3. **Joint Probability:** Joint probability refers to the probability of **two or more events occurring simultaneously**.
4. **Marginal Probability:** Marginal probability is obtained by **summing** or integrating the **joint probabilities** of an event over all possible values of the other events. It focuses on the **probability of a single event** occurring **without any consideration of any other event**.
5. **Conditional Probability:** Conditional probability builds upon the concepts of joint and marginal probabilities. It is the **probability of one event occurring given that another event has already occurred**. Understanding conditional probability is crucial for analyzing situations where events are dependent on each other.
6. **Independent Probability:** Two events are independent if the **occurrence of one event does not affect the occurrence of the other**. Their joint probability equals the product of marginal probability.
7. **Posterior Probability:** Posterior probability is the updated probability of an event after considering new evidence or information. It is calculated using **Bayes' theorem**, which **relates conditional probability to prior probability**.

Probability Rules

complement rule

$$P(A) = 1 - P(A')$$

multiplication rules (joint probability)

dependent $P(A \cap B) = P(A) * P(B|A)$

independent $P(A \cap B) = P(A) * P(B)$

mutually exclusive $P(A \cap B) = 0$

Dependent events

$$P(!X|Y) = 1 - P(X|Y)$$

$$P(B) = P(B|A) P(A) + P(B|!A)P(!A)$$

$$P(!B) = P(!B|A) P(A) + P(!B|!A)P(!A)$$

addition rules (union of events)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

mutually exclusive $P(A \cup B) = P(A) + P(B)$

conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Bayes' Theorem

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

Example

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	0.108	0.012	0.072	0.008
$\neg$ <i>cavity</i>	0.016	0.064	0.144	0.576

- Find the following:
- $P(\text{cavity})$
- $P(\text{toothache})$
- $P(\text{toothache or cavity})$
- $P(\text{cavity} | \text{toothache})$
- $P(\neg \text{cavity} | \text{toothache})$
- $P(\text{cavity} | \neg \text{toothache})$

https://www.youtube.com/watch?v=7D_4cXnj2Z0

Conditional Probability

Q1: You, your Father and Mother lineup randomly in a queue to take a memorable picture at your Convocation. Find the $P(A|B)$ such that A = Daughter on one end, B = Father in Middle.

Q2: Consider the following contingency table:

	RIGHT-HANDED	LEFT-HANDED	TOTAL
MALE	0.41	0.08	0.49
FEMALE	0.45	0.06	0.51
TOTAL	0.86	0.14	1

Find the Probability that a randomly selected person is

A: A Male given that he is right handed.

B: Right handed given that he is a male.

C: A Female given that she is left handed.

D: Are the events being a female and being left handed Independent? Justify

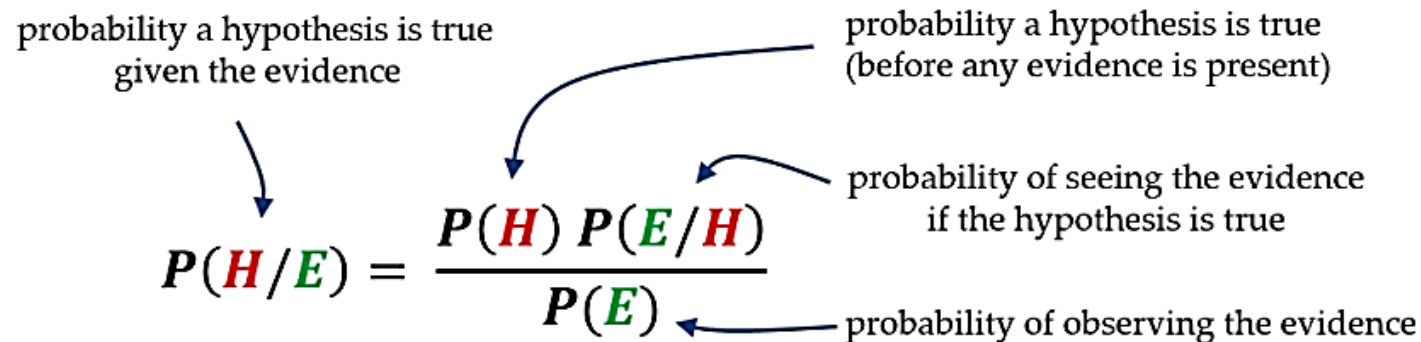
Homework

1. A doctor knows that the disease meningitis causes the patient to have a stiff neck, say, 70% of the time. The doctor also knows some unconditional facts: the prior probability that a patient has meningitis is $1/50000$, and the prior probability that any patient has a stiff neck is 1%. M =meningitis and s =stiff neck. Find the probability of meningitis given stiff neck.
2. Malaria is very common in developing countries and chances of getting malaria are 1 in 1000. 90% of the time if people have malaria, then they can have high fever. 15% of the people may have high fever given that they have other diseases such as flu, TB, etc. Find out that if you have high fever, what is the probability that you have malaria? What is the probability that you have malaria even though you do not have high fever?
3. Consider a population where 1 in 1000 individuals has a specific genetic disorder. Among individuals with this disorder, 70% exhibit a particular symptom. However, this symptom can also occur in individuals without the disorder with a probability of 5%. If a person presents with the symptom, what is the probability that they have the genetic disorder?
4. In a community, 1 in 200 people carries a rare virus. Among carriers of the virus, 75% develop mild flu-like symptoms. However, 30% of individuals who have similar symptoms may not have the virus. If a person shows flu-like symptoms, what is the probability that they are carriers of the virus?
5. Suppose a particular type of cancer is known to occur in 1 in 10,000 people. Among those with the cancer, 90% exhibit a specific biomarker. However, 15% of individuals without cancer may also have this biomarker due to other health conditions. If a person tests positive for the biomarker, what is the probability that they have the cancer?

Bayes Rule

Bayes Rule

In probability theory and statistics, Bayes' theorem (alternatively Bayes' law or Bayes' rule) describes the probability of an event, based on prior knowledge of conditions that might be related to the event.



The diagram shows the Bayes' Rule formula with arrows pointing from descriptive text to the corresponding parts of the equation:

$$P(\textcolor{red}{H}/\textcolor{green}{E}) = \frac{P(\textcolor{red}{H}) P(\textcolor{green}{E}/\textcolor{red}{H})}{P(\textcolor{green}{E})}$$

- probability a hypothesis is true given the evidence → $P(\textcolor{red}{H}/\textcolor{green}{E})$
- probability a hypothesis is true (before any evidence is present) → $P(\textcolor{red}{H})$
- probability of seeing the evidence if the hypothesis is true → $P(\textcolor{green}{E}/\textcolor{red}{H})$
- probability of observing the evidence → $P(\textcolor{green}{E})$

For example, if cancer is related to age, then, using Bayes' theorem, a person's age can be used to more accurately assess the probability that they have cancer, compared to the assessment of the probability of cancer made without knowledge of the person's age.

$$P(\textit{cause}|\textit{effect}) = \frac{P(\textit{effect}|\textit{cause})P(\textit{cause})}{P(\textit{effect})}$$

Example

Suppose there are three bowls B1, B2, B3.

- Bowl B1 has 2 red and 4 blue coins,
- Bowl B2 has 1 red and 2 blue coins, and
- Bowl B3 contains 5 red and 4 blue coins.

Suppose the probabilities for selecting the bowls is not the same but are:-

- $P(B1) = 1/3$
- $P(B2) = 1/6$
- $P(B3) = 1/2$

Now, let us compute, assuming that a red coin was drawn what will be the probability that it came from bowl B1?

In mathematical terms, we need to find out $P(B1|R)$.

According to Bayes' theorem $P(B1|R) = P(R|B1) * P(B1) / P(R)$

For that first, we need to calculate some probabilities which are:-

Probability to select a red coin i.e $P(R)$

Probability to select the bowl 1 (B1) i.e $P(B1)$ which is already given $1/3$

Probability to select a red coin from B1 i.e $P(R|B1)$

$P(R) = P(B1 \cap R) + P(B2 \cap R) + P(B3 \cap R)$, where,

- $P(B1 \cap R)$ is probability to select bowl 1 and red coin
- $P(B2 \cap R)$ is probability to select bowl 2 and red coin
- $P(B3 \cap R)$ is probability to select bowl 3 and red coin

$P(R) = P(R|B1)P(B1) + P(R|B2)P(B2) + P(R|B3)P(B3)$

$P(R) = P(\text{Number of Red coins} / \text{total number of coins in B1}) * P(\text{selecting B1}) + P(\text{Number of Red coins in B2} / \text{total number of coins in B2}) * P(\text{selecting B2}) +$

$P(\text{Number of Red coins in B3} / \text{total number of coins in B3}) * P(\text{selecting B3})$

$P(R) = 2/6 * 1/3 + 1/3 * 1/6 + 5/9 * 1/2 = 4/9$

$P(R|B1)$ = The probability of selecting a red coin given that it will be drawn from B1 is $2/6$

$P(B1)$ was given i.e. $1/3$.

By putting all the values in formula: $P(B1|R) = (2/6 * 1/3) / 4/9 = 2/8 = 0.25$

So we can say that if a red coin was drawn that it will be 25% chances that it was drawn from bowl 1 i.e. B1.

Homework

1. An online store sells two brands of smartphones, Brand A and Brand B. Brand A accounts for 60% of the sales, while Brand B accounts for the remaining 40%. The failure rate for Brand A is 5%, while for Brand B, it is 3%. If a randomly selected smartphone is found to be faulty, what is the probability that it is from Brand A?
2. A factory produces two types of light bulbs, Type X and Type Y. Type X bulbs have a defect rate of 4%, while Type Y bulbs have a defect rate of 2%. The factory produces 60% Type X bulbs and 40% Type Y bulbs. If a randomly selected defective bulb is found, what is the probability that it is of Type X?
3. A medical test for a disease is known to be 95% accurate. This disease affects 1 in 1000 people in the population. If a person tests positive for the disease, what is the probability that they actually have the disease?
4. In a city, 30% of the population are smokers, and 70% are non-smokers. Among smokers, 10% develop a particular respiratory condition, while among non-smokers, 5% develop the same condition due to other factors. If a person develops the condition, what is the probability that they are a smoker?
5. A diagnostic test for a rare condition is known to be 99% accurate. The condition affects 1 in 5000 people. If a person tests positive for the condition, what is the probability that they actually have the condition?
6. A factory produces widgets using two machines, Machine 1 and Machine 2. Machine 1 produces 60% of the widgets, while Machine 2 produces the remaining 40%. Defective widgets are found to be 2% from Machine 1 and 3% from Machine 2. If a randomly selected defective widget is found, what is the probability that it is from Machine 1?

Naïve Bayes Classifier

Naive Bayes Classifier:

- The Naive Bayes classifier calculates the probability of a class C_k given an observation $X = (x_1, x_2, \dots, x_n)$ using Bayes' theorem and the assumption of feature independence. The formula is:

$$P(C_k|X) = \frac{P(X|C_k) \cdot P(C_k)}{P(X)}$$

- Where:
 - $P(C_k|X)$ is the posterior probability of class C_k given observation X .
 - $P(X|C_k)$ is the likelihood of observing X given class C_k .
 - $P(C_k)$ is the prior probability of class C_k .
 - $P(X)$ is the probability of observing X , which acts as a normalizing factor.
- In practice, due to the independence assumption, $P(X|C_k)$ is calculated as the product of the individual conditional probabilities of each feature given the class:

$$P(X|C_k) = P(x_1|C_k) \cdot P(x_2|C_k) \cdot \dots \cdot P(x_n|C_k)$$

- The class with the highest posterior probability is predicted as the class label for the observation X .

The normalizing factor is introduced to scale probabilities so that they sum or integrate to 1, thereby ensuring that they represent valid probability distributions.

Naïve Bayes Classifier

- Naive Bayes is a kind of classifier which uses the Bayes Theorem.
- It predicts membership probabilities for each class such as the probability that given record or data point belongs to a particular class.
- The class with the highest probability is considered as the most likely class. This is also known as **Maximum A Posteriori (MAP)**.

- $\text{MAP}(H) = \max(P(H|E))$
- $\text{MAP}(H) = \max((P(E|H) * P(H)) / P(E))$
- $\text{MAP}(H) = \max(P(E|H) * P(H))$

probability a hypothesis is true given the evidence

probability a hypothesis is true (before any evidence is present)

probability of seeing the evidence if the hypothesis is true

probability of observing the evidence

$$P(H/E) = \frac{P(H) P(E/H)}{P(E)}$$

- $P(E)$ is evidence probability, and it is used to normalize the result. It remains same so, removing it won't affect.
- Naive Bayes classifier assumes that all the features are unrelated to each other. The presence or absence of a feature does not influence the presence or absence of any other feature.
- “A fruit may be considered to be an apple if it is red, round, and about 4” in diameter. Even if these features depend on each other or upon the existence of the other features, a naive Bayes classifier considers all of these properties to independently contribute to the probability that this fruit is an apple.”
- In real datasets, we test a hypothesis given multiple evidence(feature). So, calculations become complicated. To simplify the work, the feature independence approach is used to ‘uncouple’ multiple evidence and treat each as an independent one.
- $P(H | \text{Multiple Evidences}) = P(E1 | H) * P(E2 | H) \dots\dots * P(En | H) * P(H) / P(\text{Multiple Evidences})$

Example

For understanding a theoretical concept, the best procedure is to try it on an example.

Let's consider a training dataset with 1500 records and 3 classes. We presume that there are no missing values in our data. We have 3 classes associated with Animal Types:

- Parrot,
- Dog,
- Fish.

The Predictor features set consists of 4 features as

- Swim
- Wings
- Green Color
- Dangerous Teeth.

Swim, Wings, Green Color, Dangerous Teeth. All the features are categorical variables with either of the 2 values: **T(True)** or **F(False)**.

Example Continued

The below table shows a frequency table of our data. In our training data:

Swim	Wings	Green Color	Dangerous Teeth	Animal Type
50	500/500	400/500	0	Parrot
450/500	0	0	500/500	Dog
500/500	0	100/500	50/500	Fish

- Parrots have 50(10%) value for Swim, i.e., 10% parrot can swim according to our data, 500 out of 500(100%) parrots have wings, 400 out of 500(80%) parrots are Green and 0(0%) parrots have Dangerous Teeth.
- Classes with Animal type Dogs shows that 450 out of 500(90%) can swim, 0(0%) dogs have wings, 0(0%) dogs are of Green color and 500 out of 500(100%) dogs have Dangerous Teeth.
- Classes with Animal type Fishes shows that 500 out of 500(100%) can swim, 0(0%) fishes have wings, 100(20%) fishes are of Green color and 50 out of 500(10%) Fishes have Dangerous Teeth.

Example Continued

Now, it's time to work on predict classes using the Naive Bayes model.

We have taken 2 records that have values in their feature set, but the target variable needs to be predicted.

	Swim	Wings	Green	Teeth
1.	True	False	True	False
2.	True	False	True	True

- We have to predict animal type using the feature values. We have to predict whether the animal is a Dog, a Parrot or a Fish

$$P(H|\text{Multiple Evidences}) = P(E_1|H) * P(E_2|H) * \dots * P(E_n|H) * P(H) / P(\text{Multiple Evidences})$$

Let's consider the first record.

The Evidence here is Swim & Green. The Hypothesis can be an animal type to be Dog, Parrot, Fish.

Example Continued

For Hypothesis testing for the animal to be a Dog:

$$P(\text{Dog} \mid \text{Swim, Green}) = P(\text{Swim} \mid \text{Dog}) * P(\text{Green} \mid \text{Dog}) * P(\text{Dog}) / P(\text{Swim, Green})$$

$$P(\text{Dog} \mid \text{Swim, Green}) = 0.9 * 0 * 0.333 / P(\text{Swim, Green})$$

$$P(\text{Dog} \mid \text{Swim, Green}) = 0$$

For Hypothesis testing for the animal to be a Parrot:

$$P(\text{Parrot} \mid \text{Swim, Green}) = P(\text{Swim} \mid \text{Parrot}) * P(\text{Green} \mid \text{Parrot}) * P(\text{Parrot}) / P(\text{Swim, Green})$$

$$P(\text{Parrot} \mid \text{Swim, Green}) = 0.1 * 0.80 * 0.333 / P(\text{Swim, Green})$$

$$P(\text{Parrot} \mid \text{Swim, Green}) = 0.0264 / P(\text{Swim, Green})$$

For Hypothesis testing for the animal to be a Fish:

$$P(\text{Fish} \mid \text{Swim, Green}) = P(\text{Swim} \mid \text{Fish}) * P(\text{Green} \mid \text{Fish}) * P(\text{Fish}) / P(\text{Swim, Green})$$

$$P(\text{Fish} \mid \text{Swim, Green}) = 1 * 0.2 * 0.333 / P(\text{Swim, Green})$$

$$P(\text{Fish} \mid \text{Swim, Green}) = 0.0666 / P(\text{Swim, Green})$$

The denominator of all the above calculations is same i.e, $P(\text{Swim, Green})$. The value of $P(\text{Fish} \mid \text{Swim, Green})$ is greater than $P(\text{Parrot} \mid \text{Swim, Green})$.

Using Naive Bayes, we can predict that the class of this record is Fish.

Example Continued

Let's consider the second record.

The Evidence here is Swim, Green & Teeth. The Hypothesis can be an animal type to be Dog, Parrot, Fish.

For Hypothesis testing for the animal to be a Dog:

$$P(\text{Dog} \mid \text{Swim, Green, Teeth}) = P(\text{Swim} \mid \text{Dog}) * P(\text{Green} \mid \text{Dog}) * P(\text{Teeth} \mid \text{Dog}) * P(\text{Dog}) / P(\text{Swim, Green, Teeth})$$

$$P(\text{Dog} \mid \text{Swim, Green, Teeth}) = 0.9 * 0 * 1 * 0.333 / P(\text{Swim, Green, Teeth}) = 0$$

For Hypothesis testing for the animal to be a Parrot:

$$P(\text{Parrot} \mid \text{Swim, Green, Teeth}) = P(\text{Swim} \mid \text{Parrot}) * P(\text{Green} \mid \text{Parrot}) * P(\text{Teeth} \mid \text{Parrot}) * P(\text{Parrot}) / P(\text{Swim, Green, Teeth})$$

$$P(\text{Parrot} \mid \text{Swim, Green, Teeth}) = 0.1 * 0.80 * 0 * 0.333 / P(\text{Swim, Green, Teeth})$$

$$P(\text{Parrot} \mid \text{Swim, Green, Teeth}) = 0$$

For Hypothesis testing for the animal to be a Fish:

$$P(\text{Fish} \mid \text{Swim, Green, Teeth}) = P(\text{Swim} \mid \text{Fish}) * P(\text{Green} \mid \text{Fish}) * P(\text{Teeth} \mid \text{Fish}) * P(\text{Fish}) / P(\text{Swim, Green, Teeth})$$

$$P(\text{Fish} \mid \text{Swim, Green, Teeth}) = 1 * 0.2 * 0.1 * 0.333 / P(\text{Swim, Green, Teeth})$$

$$P(\text{Fish} \mid \text{Swim, Green, Teeth}) = 0.00666 / P(\text{Swim, Green, Teeth})$$

The value of $P(\text{Fish} \mid \text{Swim, Green, Teeth})$ is the only positive value greater than 0.

Using Naive Bayes, we can predict that the class of this record is **Fish**.

Bayesian Networks

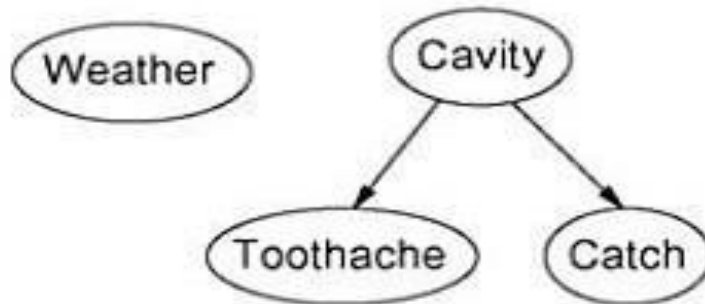
Motivations

- ❑ Full joint probability distribution can answer any question but can become intractably large as number of variable increases
- ❑ Specifying probabilities for atomic events can be difficult, e.g., large set of data, statistical estimates, etc.
- ❑ Independence and conditional independence reduce the probabilities needed for full joint probability distribution.
 - Full joint probability distribution describes the **probabilities of all possible combinations of values** for a set of random variables.
 - Independence refers to the **absence of a probabilistic relationship** between variables.
 - Conditional independence allows for independence under certain conditions, given the value of another variable.
 - Two random variables X and Y are conditionally independent given another random variable Z if and only if their joint probability distribution factorizes into the product of their conditional probability distributions given Z: $P(X,Y|Z)=P(X|Z) \cdot P(Y|Z)$
 - In words, given the value of Z, knowing the value of one variable provides no additional information about the value of the other variable beyond what is already provided by Z.

Bayesian networks

- ❑ A simple, graphical notation for conditional independence assertions and hence for compact specification of full joint distributions
- ❑ A directed, acyclic graph (DAG)
- ❑ A set of nodes, one per variable (discrete or continuous)
- ❑ A set of directed links (arrows) connects pairs of nodes. X is a parent of Y if there is an arrow (direct influence) from node X to node Y .
- ❑ Each node X_i has a conditional probability distribution $P(X_i | Parents(X_i))$ that quantifies the effect of the parents on the node.
- ❑ Combinations of the topology and the conditional distributions specify (implicitly) the full joint distribution for all the variables.

Topology of network encodes conditional independence assertions:

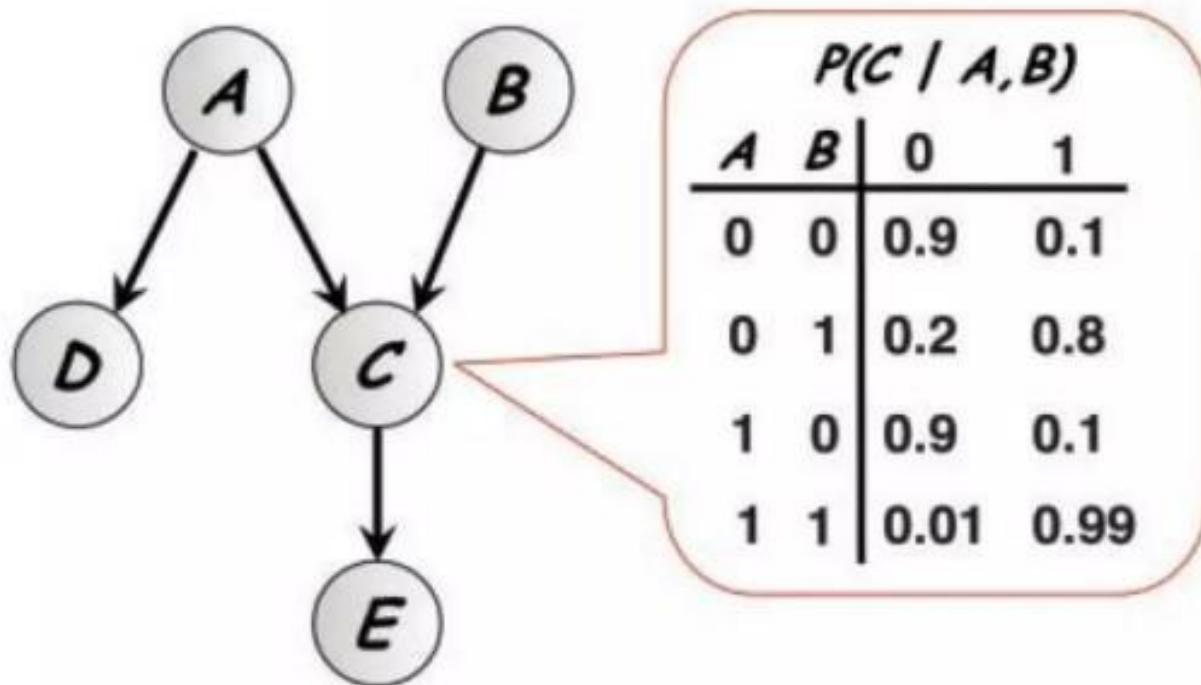


✓ Causes
should
be
parents
that of
effects

Weather is independent of the other variables

Toothache and *Catch* are conditionally independent given *Cavity*

A **Bayesian network** is a graphical model for probabilistic relationships among a set of variables



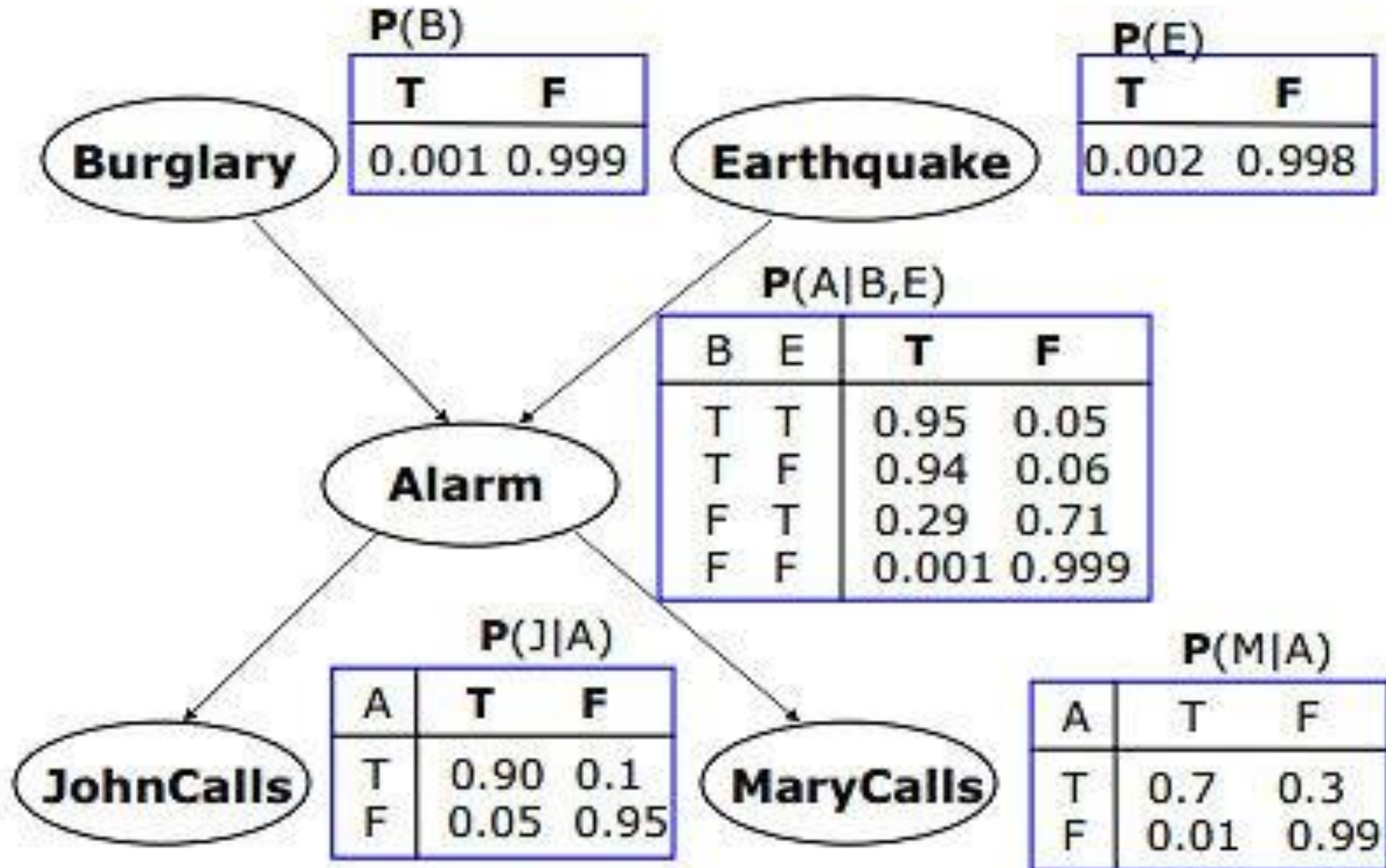
$$P(A, B, C, D, E) = P(A)P(B)P(C | A, B)P(D | A)P(E | C)$$

Example: Burglar alarm system

- I have a burglar alarm installed at home
 - It is fairly reliable at detecting a burglary, but also responds on occasion to minor earth quakes.
- I also have two neighbors, John and Mary
 - They have promised to call me at work when they hear the alarm
 - John always calls when he hears the alarm, but sometimes confuses the telephone ringing with the alarm and calls then, too.
 - Mary likes rather loud music and sometimes misses the alarm altogether.
- Bayesian networks variables:
 - *Burglar, Earthquake, Alarm, JohnCalls, MaryCalls*

Bayesian belief network.

Variable Order = [John, Mary, Alarm, Burglar, Earthquake]



Compactness of Bayesian networks

CPT: Conditional Probability Table

A CPT for Boolean X_i with k Boolean parents has 2^k rows for the combinations of parent values

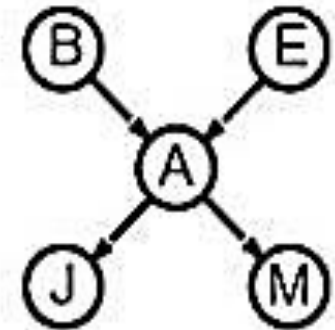
Each row requires one number p for $X_i = \text{true}$ (the number for $X_i = \text{false}$ is just $1 - p$)

If each variable has no more than k parents, the complete network requires $O(n \cdot 2^k)$ numbers

I.e., grows linearly with n , vs. $O(2^n)$ for the full joint distribution

For burglary net, $1 + 1 + 4 + 2 + 2 = 10$ numbers (vs. $2^5 - 1 = 31$)

- By decomposing the joint distribution into smaller conditional distributions, we reduce the number of parameters that need to be estimated.
- **Instead of estimating the entire joint distribution, we only need to estimate the conditional probabilities associated with each node given its parents.**



Example 1

The joint probability $P(A,B)$ is calculated based on the conditional probability of B given A (or vice versa), multiplied by the probability of A occurring.

$$P(A) = \sum_{b \in \{B, !B\}} \sum_{e \in \{E, !E\}} P(A, b, e)$$

Recall from conditional probability: $P(A,B)=P(A|B) \cdot P(B)$

So, $P(A,B,E)=P(A|B,E) \cdot P(B,E)$

Which means:

$$P(A) = P(A|B,E)P(B,E) + P(A|B,!E)P(B,!E) + P(A|!B,E)P(!B,E) + P(A|!B,!E)P(!B,!E)$$

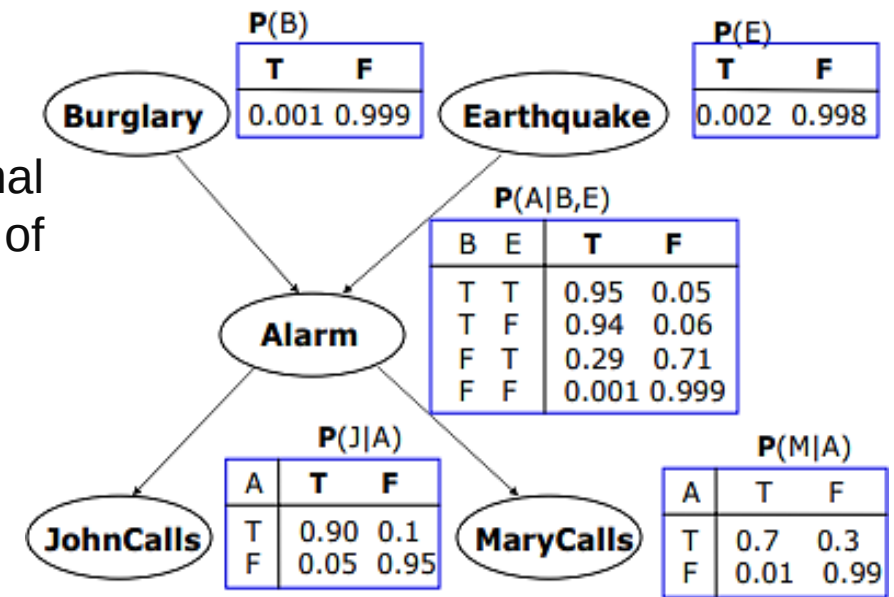
This is because there are 4 ways in which A can happen.

A can happen:

1. if events B and E happen,
2. if B happens and not E,
3. if E happens and not B, or
4. if neither of them happens.

Working that out (and assuming B and E are independent), we have:

$$P(A)=(0.95*0.001*0.002)+(0.94*0.001*0.998)+(0.29*0.999*0.002)+(0.001*0.999*0.998)=0.002516442$$



Global Semantics

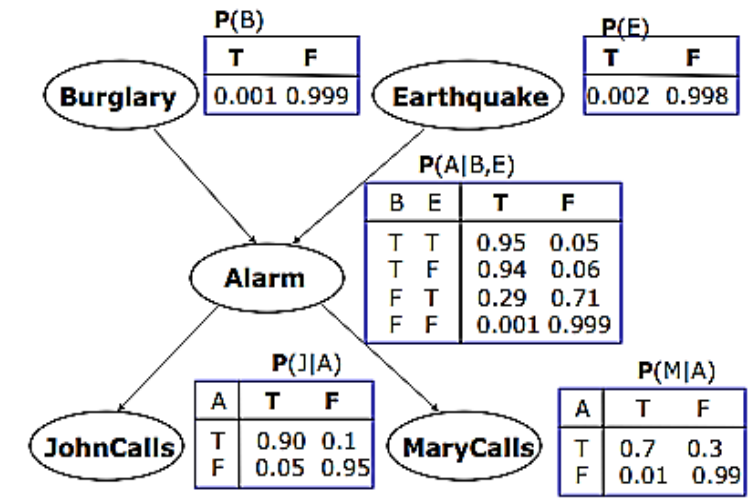
1. What is the probability that the alarm has sounded but neither a burglary nor an earthquake has occurred, and both John and Merry call?

"Global" semantics defines the full joint distribution as the product of the local conditional distributions:

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$

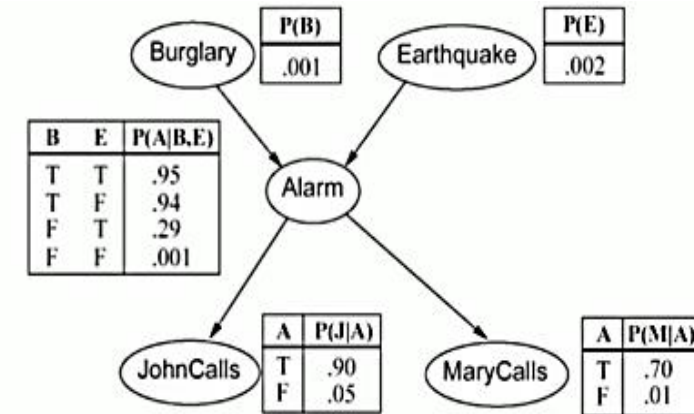
e.g., $P(j \wedge m \wedge a \wedge \neg b \wedge \neg e)$

$$\begin{aligned} &= P(j|a)P(m|a)P(a|\neg b, \neg e)P(\neg b)P(\neg e) \\ &= 0.9 \times 0.7 \times 0.001 \times 0.999 \times 0.998 \\ &\approx 0.00063 \end{aligned}$$



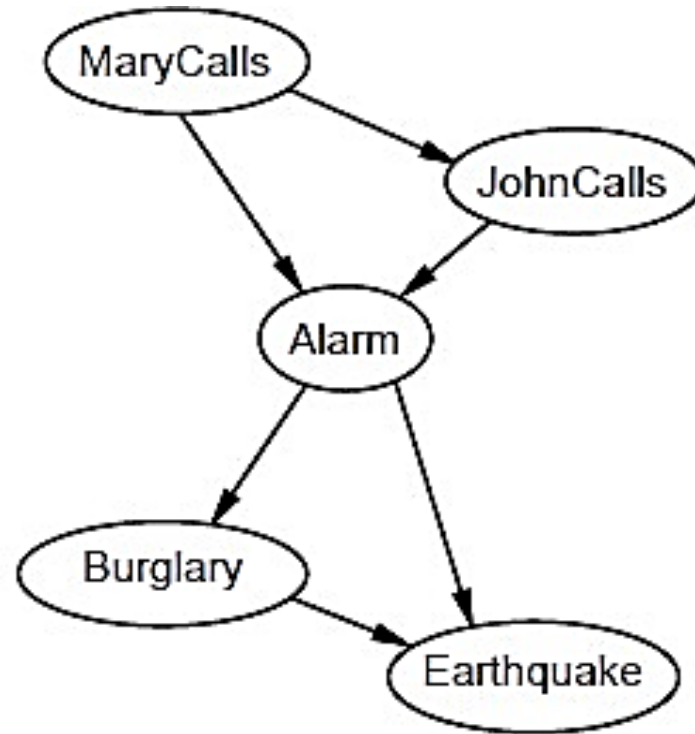
2. What is the probability that John call?

Solution:



$$\begin{aligned} P(j) &= P(j | a) P(a) + P(j | \neg a) P(\neg a) \\ &= P(j|a) \{P(a|b,e) * P(b,e) + P(a|\neg b,e) * P(\neg b,e) + P(a|b,\neg e) * P(b,\neg e) + P(a|\neg b,\neg e) * P(\neg b,\neg e)\} \\ &\quad + P(j|\neg a) \{P(\neg a|b,e) * P(b,e) + P(\neg a|\neg b,e) * P(\neg b,e) + P(\neg a|b,\neg e) * P(b,\neg e) + P(\neg a|\neg b,\neg e) * P(\neg b,\neg e)\} \\ &= 0.90 * 0.00252 + 0.05 * 0.9974 = 0.0521 \end{aligned}$$

Example



Deciding conditional independence is hard in noncausal directions

(Causal models and conditional independence seem hardwired for humans!)

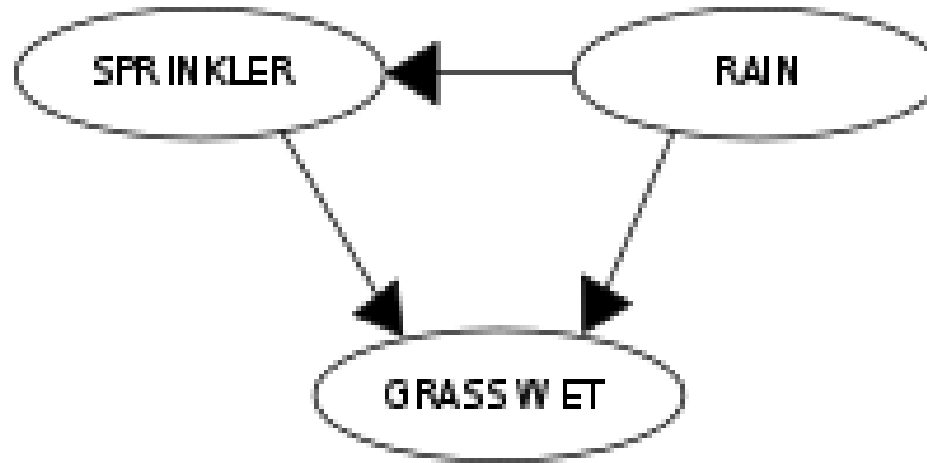
Assessing conditional probabilities is hard in noncausal directions

Network is less compact: $1 + 2 + 4 + 2 + 4 = 13$ numbers needed

Example 2

Two events can cause grass to be wet: an active sprinkler or rain. Rain has a direct effect on the use of the sprinkler (namely that when it rains, the sprinkler usually is not active). This situation can be modeled with a Bayesian network. Each variable has two possible values, T (for true) and F (for false).

RAIN	SPRINKLER	
	T	F
F	0.4	0.6
T	0.01	0.99



	RAIN	
	T	F
	0.2	0.8



SPRINKLER	RAIN	GRASS WET	
		T	F
F	F	0.0	1.0
F	T	0.8	0.2
T	F	0.9	0.1
T	T	0.99	0.01

What is the probability that it is raining, given the grass is wet?

What is the probability that it is raining, given the grass is wet?

- We can then calculate, for example:

$$P(\text{it is raining} \mid \text{grass is wet}) = \frac{P(\text{it is raining AND grass is wet})}{P(\text{grass is wet})}$$

$$= \frac{\sum_{\text{sprinkler} \in \{T, F\}} P(\text{grass is wet} = T \text{ AND sprinkler AND raining} = T)}{\sum_{\text{sprinkler} \in \{T, F\}, \text{ raining} \in \{T, F\}} P(\text{grass is wet} = T \text{ AND sprinkler AND raining})}$$

The joint probability Distribution formula is

$$\Pr(G, S, R) = \Pr(G|S, R) \Pr(S|R) \Pr(R)$$

where G = Grass wet (true/false), S = Sprinkler turned on (true/false), and R = Raining (true/false).

What is the probability that it is raining, given the grass is wet?

The joint probability distribution formula is

$$\Pr(G, S, R) = \Pr(G|S, R) \Pr(S|R) \Pr(R)$$

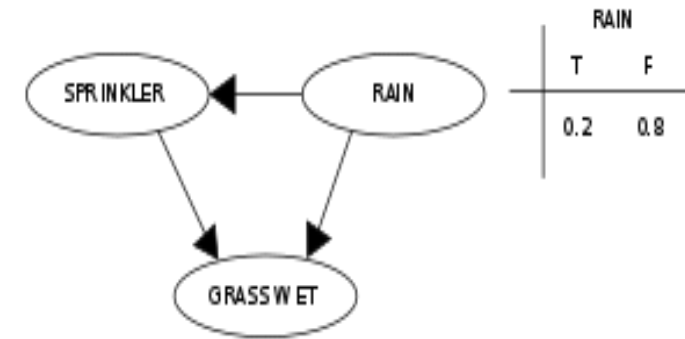
Where,

G = Grass wet (true/false), S = Sprinkler turned on (true/false), and R = Raining (true/false).

By using the conditional probability formula and summing over all random variables:

$$\Pr(R = T|G = T) = \frac{\Pr(G = T, R = T)}{\Pr(G = T)} = \frac{\sum_{S \in \{T, F\}} \Pr(G = T, S, R = T)}{\sum_{S, R \in \{T, F\}} \Pr(G = T, S, R)}$$

RAIN	SPRINKLER	
	T	F
F	0.4	0.6
T	0.01	0.99



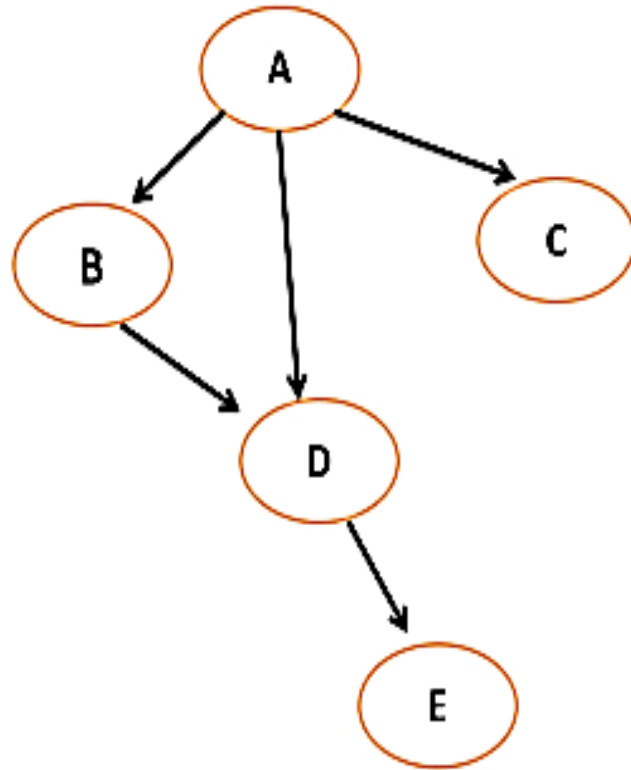
SPRINKLER	RAIN	GRASS WET	
		T	F
F	F	0.0	1.0
F	T	0.8	0.2
T	F	0.9	0.1
T	T	0.99	0.01

Using the expansion for the joint probability function $\Pr(G, S, R)$ and the conditional probabilities from CPT

$$\begin{aligned} \Pr(G = T, S = T, R = T) &= \Pr(G = T|S = T, R = T) \Pr(S = T|R = T) \Pr(R = T) \\ &= 0.99 \times 0.01 \times 0.2 \\ &= 0.00198. \end{aligned}$$

$$\Pr(R = T|G = T) = \frac{0.00198_{TTT} + 0.1584_{TFT}}{0.00198_{TTT} + 0.288_{TTF} + 0.1584_{TFT} + 0.0_{TFF}} = \frac{891}{2491} \approx 35.77\%.$$

Example 3



$$P(A=ON)=0.6$$

$$P(B=ON|A)=\begin{cases} 0.1, & A=OFF \\ 0.95, & A=ON \end{cases}$$

$$P(C=ON|A)=\begin{cases} 0.8, & A=OFF \\ 0.5, & A=ON \end{cases}$$

$$P(D=ON|A,B)=\begin{cases} 0.1 & A=OFF, B=OFF \\ 0.9 & A=ON, B=OFF \\ 0.3 & A=OFF, B=ON \\ 0.95 & A=ON, B=ON \end{cases}$$

$$P(E=ON|D)=\begin{cases} 0.8, & D=OFF \\ 0.1, & D=ON \end{cases}$$

1. What is $P(A=on, B=on, C=on, D=on, E=on)$?
2. What is $P(E=on | A=on)$?
3. What is $P(A=on | E=on)$?

• **Solution:**

1. $P(A=on, B=on, C=on, D=on, E=on)$

- $P(A=ON)P(B=ON|A=ON)P(C=ON|A=ON)P(D=ON|A=ON, B=ON)P(E=ON|D=ON) = (0.6)(0.95)(0.5)(0.95)(0.1) = 0.0271$

2. $P(E=on | A=on)$

- $P(E=on | A=on) = P(E=on, A=on) / P(A=on)$

- $P(E=on | A=on) = P(B|A=ON) P(D|A=ON, B) P(E=ON|D)$

- In this case, B and D will have a different combo of on and off

- $P(E=on | A=on) = (0.1*0.9*0.05) + (0.1*0.95*0.95) + (0.8*0.05*0.95) + (0.8*0.1*0.05)$

- $P(E=on | A=on) = 0.13675$

3. $P(A=on | E=on)$

- $P(A=on | E=on) = P(A=on, E=on) / P(E=on)$

- $P(A=on | E=on) = 0.2227$

- What is $P(A, B, C, D, E)$ if they are mutually independent?
- What is $P(A, B, C, D, E)$ as per independence assumptions of the network provided?

<https://www.youtube.com/watch?v=F0L4oQKRrpE>

<https://www.youtube.com/watch?v=MqZyv4qiyJ8>